Customer Churn Analysis Report

1. Project Overview

Objective:
Analyze telecom customer churn to identify high-risk segments and provide actionable insights for retention strategies. This project combines **Python**, **SQL**, and **Power BI** to perform end-to-end data analysis and dashboard visualization.

Goals:

- Determine overall churn rate and contributing factors.
- Analyze customer behavior across demographics, contract types, and services.
- Build a dynamic dashboard for business insights and decision-making.

2. Tools & Technologies

Tool / Language	Purpose
Python (pandas, numpy, matplotlib, seaborn)	Data cleaning, EDA, and visualization
SQL	Aggregation, queries, and churn metric calculation
Power BI	Interactive dashboards and KPI visualization
DAX	Metrics and calculations for dashboards

3. Dataset Overview

Dataset Summary:

- **Records:** 7,043 customers

- **Columns:** 21 (demographics, account info, services, financials, churn)

Key Features:

- Demographics: Gender, Senior Citizen, Partner, Dependents
- Account Info: Tenure, Contract Type, Payment Method, Paperless Billing
- Services: Phone Service, Internet Service, Tech Support, Streaming
- Financials: Monthly Charges, Total Charges
- Target Variable: Churn (Yes / No)

Data Quality Measures:

- Missing values in total_charges (11 rows) replaced with 0.
- Added numeric churn_count column for aggregation.
- Removed duplicate customer records.

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4. Exploratory Data Analysis (EDA) Insights

- **Total churned customers:** 1,869 (~26.5%)
- **Churn Patterns:**
  - High churn among low-tenure customers and month-to-month contracts
  - Electronic check payment users have higher churn rates
  - Fiber optic users on month-to-month plans show higher churn
- **Demographics Observations:**
  - Gender distribution roughly equal; slight variation in churn
  - Senior citizens show lower churn rates
- **Financial Observations:**
  - Higher monthly charges correlate with higher churn for shorter tenure customers

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5. SQL Analysis Highlights

- **Total Customers:** 7,043
- **Churn Rate:** 26.5%
- **Churn by Gender:** Slight difference observed between male and female customers
- **Churn by Contract:** Month-to-month contracts are highest risk
- **Churn by Payment Method:** Electronic Check users are highest risk
- **High-Risk Segment:** Customers with tenure <6 months and monthly charges > 70

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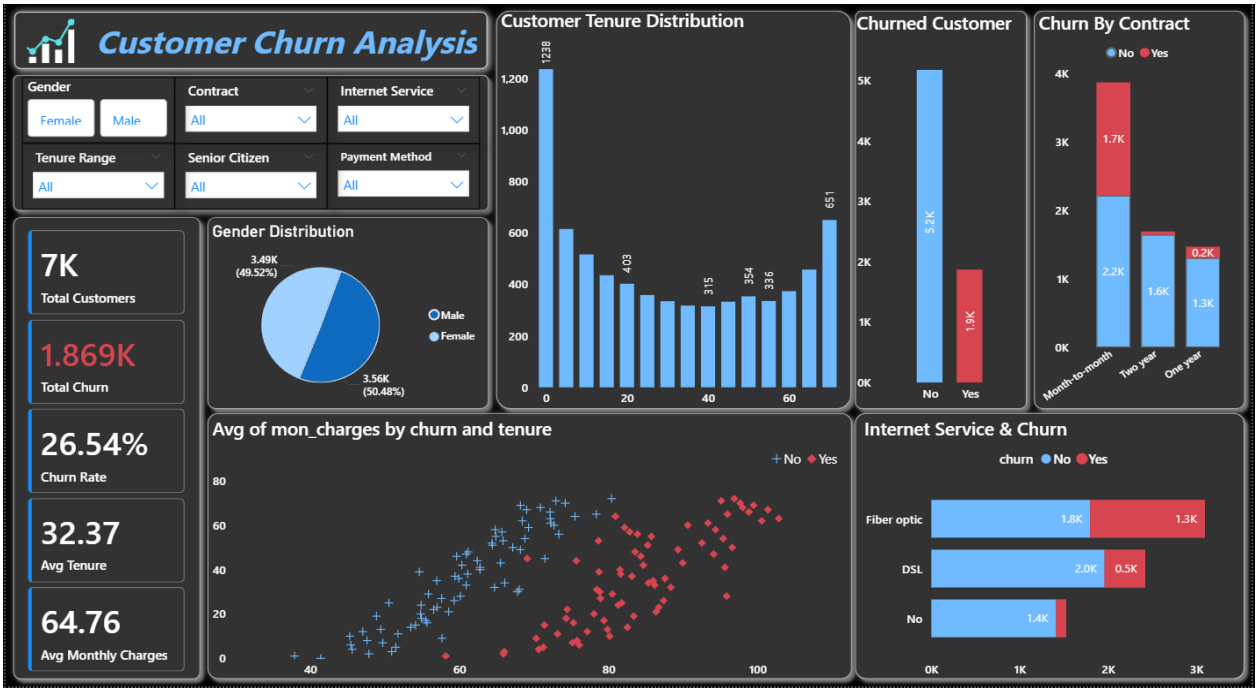
6. Power BI Dashboard Metrics

Key Metrics (DAX):

- Total Customers: Count of all records
- Total Churn: Sum of churn_count
- Churn Rate (%): Total Churn / Total Customers
- Average Monthly Charges
- Average Tenure
- Insight: Fiber optic users at higher risk



7.Recommendations

- Target month-to-month contract customers for retention campaigns.
- Offer promotions or discounts to high monthly charge customers.
- Focus customer support and engagement for fiber optic subscribers.
- Encourage electronic check users to adopt automated payment methods to reduce churn risk.

8. Conclusion

This analysis identifies high-risk churn segments and actionable strategies to improve customer retention. The combination of **Python, SQL, and Power BI** demonstrates end-to-end analytics capability and provides clear insights for business decision-making.